

FLIGHT BOOKING & TRAVEL DATA ANALYSIS

1. Project Overview and Objective

The flight booking industry generates large volumes of data related to airlines, flights, routes, ticket prices, travel classes, and customer booking behaviour. However, raw booking data can contain data quality issues such as missing values, duplicate records, inconsistent formats, spelling variations, mixed data types, and invalid entries. These issues can affect the accuracy of analysis and business decision-making.

The objective of this project is to clean, transform, and analyze a flight booking and travel dataset using Excel and Power BI. The project focuses on improving data quality, combining related airline, booking, flight, and route information, and creating an interactive dashboard to generate meaningful business insights.

2. Data Sources

The flight booking dataset used in this project was obtained from Hugging Face, an online platform that provides datasets for machine learning and data analysis.

The original dataset was reviewed and reduced to 1,200 relevant booking records using Claude AI to make the dataset suitable for analysis while retaining the required information for studying airlines, routes, pricing, booking behaviour, and flight characteristics.

The prepared dataset contains four main tables:

- Airlines – airline-level booking and pricing information
- Bookings – individual booking records and passenger booking details
- Flights – flight schedules, cities, stops, and duration
- Routes – route-level booking, pricing, duration, and airline information

Data Source: Hugging Face

Domain: Travel & Aviation

Tools Used: Microsoft Excel, Power Query, Power BI

3. Problem Statement

- To analyze booking performance across different airlines and identify airlines with higher booking demand.
 - To study booking patterns across travel classes and understand Economy and Business class preferences.
 - To compare ticket prices across airlines, routes, travel classes, and price categories.
 - To analyze customer booking behaviour based on the number of days before travel.
 - To examine the relationship between flight duration, number of stops, and ticket prices.
 - To identify frequently booked routes and understand route-level flight and airline coverage.
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4. Attribute (Column / Features) Details

4.1 Airlines Table

Attribute Name	Data Type	Description
Airline_Id	String	Unique identifier for each airline
Airline_Name	String	Name of the airline
Total_Bookings	Numeric	Total number of bookings for the airline
Routes_Covered	Numeric	Number of routes covered by the airline
Avg_Price	Numeric / Currency	Average ticket price
Min_Price	Numeric / Currency	Minimum observed ticket price
Max_Price	Numeric / Currency	Maximum observed ticket price
Economy_Bookings	Numeric	Number of Economy class bookings
Business_Bookings	Numeric	Number of Business class bookings

4.2 Bookings Table

Attribute Name	Data Type	Description
Booking_Id	String	Unique identifier for each booking
Flight_Id	String	Identifier linking the booking to a flight
Route_Id	String	Identifier linking the booking to a route
Airline_Id	String	Identifier linking the booking to an airline
Airline_Name	String	Name of the airline
Class	Categorical	Travel class selected by the customer
Stops	Categorical / Numeric	Number of stops in the flight
Duration	Numeric / Text	Flight duration
Days_Left	Numeric	Number of days between booking and travel
Price	Numeric / Currency	Ticket price

4.3 Flights Table

Attribute Name	Data Type	Description
Flight_Id	String	Unique identifier for each flight
Airline_Id	String	Identifier linking the flight to an airline
Flight_Code	String	Flight code
Route_Id	String	Identifier linking the flight to a route
Source_City	Categorical	Departure city
Destination_City	Categorical	Arrival city
Departure_Time	Time / Text	Scheduled departure time
Arrival_Time	Time / Text	Scheduled arrival time
Stops	Categorical / Numeric	Number of stops
Duration	Numeric / Text	Flight duration

4.4 Routes Table

Attribute Name	Data Type	Description
Route_Id	String	Unique identifier for each route
Source_City	Categorical	Starting city of the route
Destination_City	Categorical	Destination city
Total_Bookings	Numeric	Total bookings for the route
Unique_Flights	Numeric	Number of unique flights operating on the route
Airlines_Serving	Numeric	Number of airlines serving the route
Avg_Price	Numeric / Currency	Average ticket price for the route
Min_Price	Numeric / Currency	Minimum ticket price
Max_Price	Numeric / Currency	Maximum ticket price
Avg_Duration_Hrs	Numeric	Average flight duration in hours
Economy_Bookings	Numeric	Economy class bookings
Business_Bookings	Numeric	Business class bookings

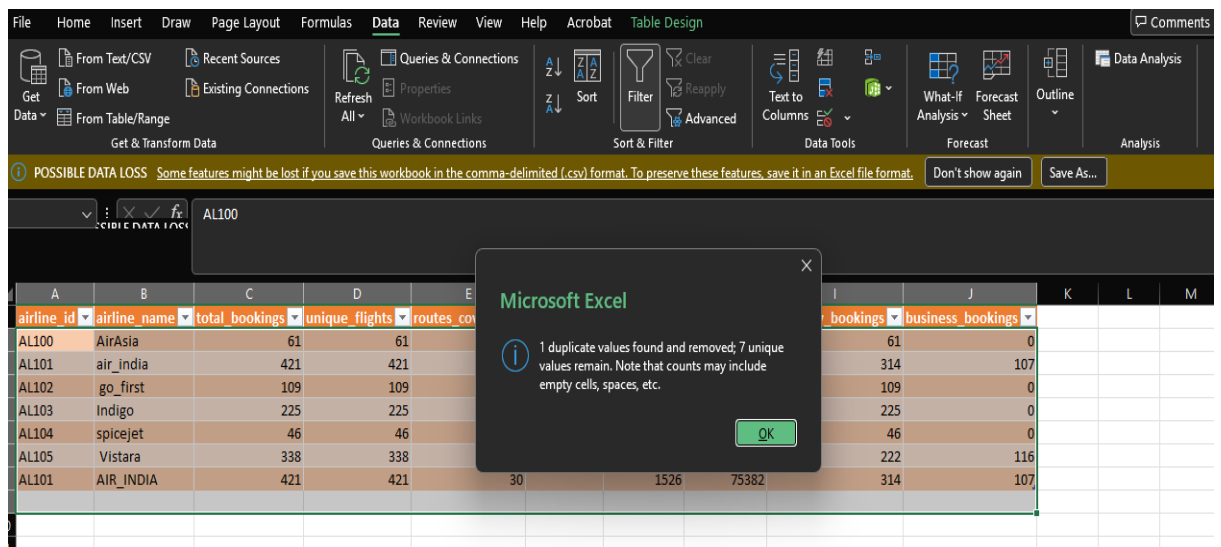
5. Tools & Technologies

- Excel: Data cleaning, transformation, formulas, calculated fields, validation, and Pivot Tables.
- Power Query: Data preparation, standardization, missing-value handling, and transformation.
- Power BI: Data modelling, DAX calculations, visualization, interactive filtering, and dashboard creation.

6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

- Data Cleaning & Transformation: Removed duplicate booking records, handled missing values, standardized formats, corrected inconsistent city and airline names, and cleaned currency values.



- **Missing Value Handling:** Missing Price values were handled using the median value. Missing Stops and Duration values in the Bookings data were completed using the corresponding Flight information.
- **Data Standardization:** Standardized airline names, city names, Stops values, Duration values, date/time-related fields, and currency formatting.

POSSIBLE DATA LOSS Some features might be lost if you save this workbook in the comma-delimited (.csv) format. To preserve these features, save it in an Excel file format. Don't show again Save As...

3 : X ✓ fx =TRIM(PROPER([@[airline_name]]))

airline_id	airline_name	total_bookings	unique_flights	routes covered	avg_price	min_price	max_price	economy_bookings	business_bookings	Names
AL100	AirAsia	61	61	27	3,465	1443	7484	61	0	Airasia
AL101	air_india	421	421	30	19,084	1526	75382	314	107	Air_India
AL102	go_first	109	109	26	6,255	2124	-	109	0	Go_First
AL103	Indigo	225	225	30	NA		24561	225	0	Indigo
AL104	spicejet	46	46	22	6,045	1566	N/A	46	0	Spicejet
AL105	Vistara	338	338	30	26599	1776	110936	222	116	Vistara
AL101	AIR_INDIA	421	421	30		1526	75382	314	107	Air_India

- **Duplicate Handling:** Duplicate booking records were identified and removed while preserving valid repeated Flight IDs and Route IDs.
- **Data Type Conversion:** Corrected numeric, currency, text, and date/time data types for accurate analysis.

File Home Insert Draw Page Layout Formulas Data Review View Help Acrobat Table Design

Clipboard Font Alignment

Price

Booking_id	Flight_id	Route_id	Airline_id	Airline_Name	Class	Stops	Booking_Window	Price
BK1000	FLT10002	RT1	AL100	AirAsia	Economy	0	14 Short Notice	₹ 5,956.00
BK1001	FLT10006	RT1	AL105	Vistara	Business	0	15 Short Notice	₹ 28,872.00
BK1002	FLT10010	RT1	AL102	Go_First	Economy	0	30 Moderate	₹ 2,410.00
BK1003	FLT10030	RT1	AL102	Go_First	Economy	1	38 Advance	₹ 7,907.00
BK1004	FLT10070	RT1	AL105	Vistara	Business	1	19 Moderate	₹ 41,281.00
BK1005	FLT10073	RT1	AL105	Vistara	Business	1	Last Minute	₹ 70,793.00
BK1006	FLT10077	RT1	AL105	Vistara	Economy	1	2 Last Minute	₹ 17,348.00
BK1007	FLT10078	RT1	AL105	Vistara	Economy	1	2 Last Minute	₹ 15,353.00
BK1008	FLT10079	RT1	AL103	Indigo	Economy	1	38 Advance	₹ 3,054.00
BK1009	FLT10081	RT1	AL103	Indigo	Economy	1	39 Advance	₹ 3,631.00

General No specific format

123 Number Price

Currency Price

Accounting Price

Short Date Price

Long Date Price

Time Price

Percentage Price

- Data Validation: Verified booking totals, airline totals, route totals, class totals, and key numerical fields.
- Calculated Fields: Created Booking_Window, Duration_Category, Price_Category, and Route fields for analytical purposes.

8 $\text{=CONCATENATE(E8,"->",F8)}$

Flight Id	Airline Id	Flight Code	Route Id	Source City	Destination City	Departure Time	Arrival Time	Stops	Duration	Route
FLT22651	AL105	UK-811	RT1	Delhi	Mumbai	Morning	Night	1	07:55:12	Delhi -> Mumbai
FLT10081	AL103	6E-2102	RT1	Delhi	Mumbai	Morning	Afternoon	1	01:55:12	Delhi -> Mumbai
FLT10140	AL101	AI-544	RT1	Delhi	Mumbai	Evening	Night	1	16:04:48	Delhi -> Mumbai
FLT10079	AL103	6E-184	RT1	Delhi	Mumbai	Morning	Afternoon	1	04:04:48	Delhi -> Mumbai
FLT10271	AL103	6E-2137	RT1	Delhi	Mumbai	Night	Night	0	04:04:48	Delhi -> Mumbai
FLT10229	AL105	UK-835	RT1	Delhi	Mumbai	Evening	Morning	1	01:55:12	Delhi -> Mumbai
FLT10429	AL102	G8-713	RT1	Delhi	Mumbai	Night	Night	1	22:04:48	Delhi -> Mumbai
FLT10400	AL102	G8-119	RT1	Delhi	Mumbai	Evening	Morning	1	18:00:00	Delhi -> Mumbai
FLT10082	AL101	AI-801	RT1	Delhi	Mumbai	Evening	Night	1	07:55:12	Delhi -> Mumbai
FLT10305	AL102	G8-717	RT1	Delhi	Mumbai	Evening	Night	1	07:55:12	Delhi -> Mumbai
FLT10185	AL101	AI-504	RT1	Delhi	Mumbai	Night	Night	1	19:55:12	Delhi -> Mumbai
FLT10010	AL102	G8-387	RT1	Delhi	Mumbai	Afternoon	Evening	0	06:00:00	Delhi -> Mumbai

- Pivot Tables: Created Pivot Tables for airline performance, price analysis, route analysis, booking-window analysis, and class-wise price analysis.

G17 Row Labels

Flight Booking and Travel Data Analysis

Airline Class Performance				Airline-wise Price Analysis			
Row Labels	Total Economy Booking	Total Business Booking	Total Bookings	Row Labels	Avg Price	Min Price	Max Price
Air India	314	107	421	Air India	18373.67933	1526	75382
AirAsia	61	0	61	AirAsia	3990.52459	1443	7484
Go First	109	0	109	Go First	6339.715596	2124	31773
Indigo	225	0	225	Indigo	5847.977778	1105	24561
Spicejet	46	0	46	Spicejet	6134.978261	1566	27769
Vistara	222	116	338	Vistara	25590.51775	1776	110936
Grand Total	977	223	1200	Grand Total	11036.23222	1105	110936

Route-wise Booking Analysis				Booking Window Analysis			
Row Labels	Average of Duration_Hr	Average of Price	Count of Booking Id	Row Labels	Count of Booking Id	Average of Price	
Bengaluru -> C	18.53525	14582.425	40	Advance	602	₹ 13,952	
Bengaluru -> D	11.719	11995.55	40	Last Minute	313	₹ 20,829	
Bengaluru -> E	15.93575	16989.55	40	Moderate	165	₹ 12,277	
Bengaluru -> F	13.79175	16061.4	40	Short Notice	120	₹ 16,413	
Bengaluru -> G	13.833	19002.775	40	Grand Total	1200	₹ 15,768	

Descriptive Statistics

Total Bookings	Routes Covered	Avg Price	Min Price	Max Price
Mean	200	Mean 27.5	Mean 11036.23222	Mean 1590
Standard Error	63.20021097	Standard Error 1.31021627	Standard Error 3601.63843	Standard Error 139.195067
Median	167	Median 28.5	Median 6237.346929	Median 1546
Mode	#N/A	Mode 30	Mode #N/A	Mode #N/A
Standard Deviation	154.8082685	Standard Deviation 3.20936131	Standard Deviation 8822.176391	Standard Deviation 340.956889
Sample Variance	23965.6	Sample Variance 10.3	Sample Variance 77830796.27	Sample Variance 116251.6
Kurtosis	-1.676163792	Kurtosis 0.59477802	Kurtosis -0.22870615	Kurtosis 0.91478223
Skewness	0.513151946	Skewness -1.11627328	Skewness 1.205119058	Skewness 0.31161618
Range	375	Range 8	Range 21659.99316	Range 1019
Minimum	46	Minimum 22	Minimum 3930.52459	Minimum 1105
Maximum	421	Maximum 30	Maximum 25590.51775	Maximum 2124
Sum	1200	Sum 165	Sum 66217.39331	Sum 9540
Count	6	Count 6	Count 6	Count 6

Descriptive Statistics
Input: Input Range: \$C\$5:\$H\$7
Grouped By: Columns
Labels in first row
Output options: Output Range: \$A\$2
New Worksheet By: New Workbook
Summary statistics: Confidence Level for Mean: 95 %
Kth Largest: 1
Kth Smallest: 1

7. Data Modelling and DAX (Power BI)

Data Model:

Established relationships between the analytical table and supporting tables.

The main relationships include:

- Airlines → Analysis
- Routes → Analysis
- Flights → Analysis

The Bookings table was kept separate from the Analysis table to avoid duplicate counting and relationship ambiguity.

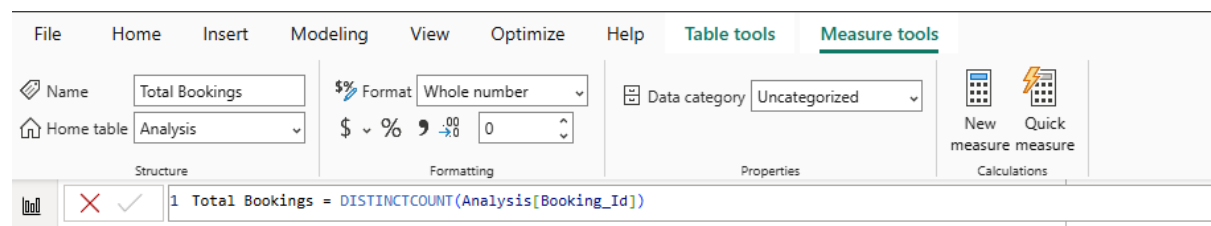
+ New relationship		Autodetect	Edit	Delete	Filter
☐ From: table (column) ↑	Relationship	To: table (column)	Status		
☐ Anal Analysis (Flight_Id)	* ← 1	Airlines (Airline_Id)	Active	...	
☐ Analysis (Flight_Id)	* ↔ 1	Flights (Flight_Id)	Active	...	
☐ Analysis (Route_Id)	* ← 1	Routes (Route_Id)	Active	...	

DAX Measures

The following measures were implemented:

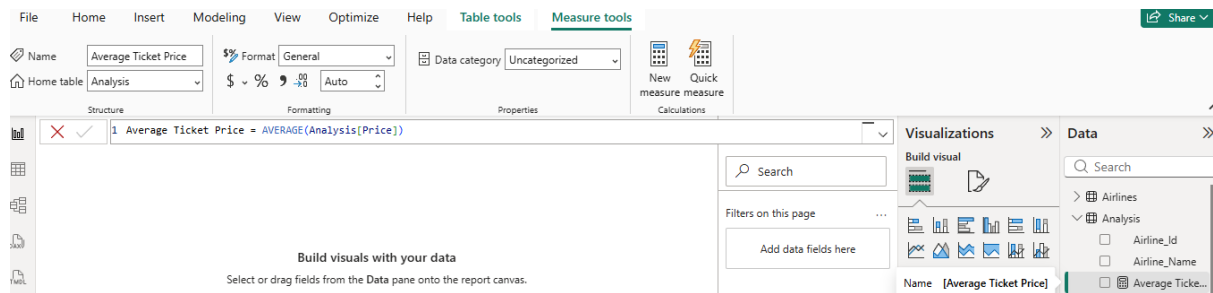
Total Bookings

Total Bookings = DISTINCTCOUNT(Analysis[Booking_Id])



Average Ticket Price

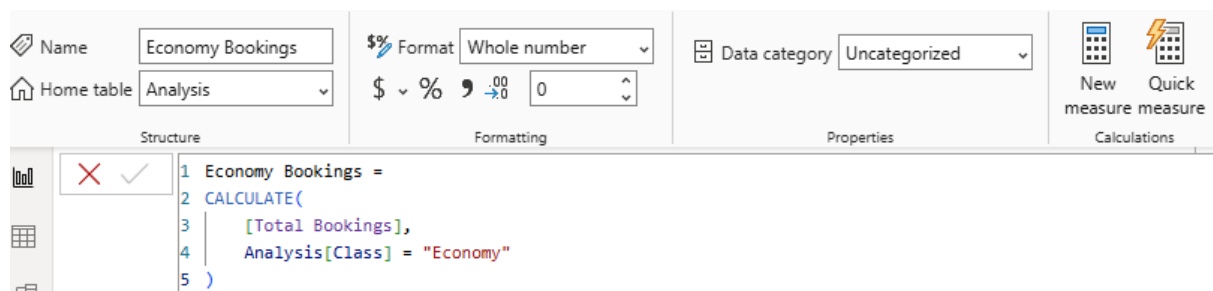
Average Ticket Price = AVERAGE(Analysis[Price])



Economy Bookings

Economy Bookings =

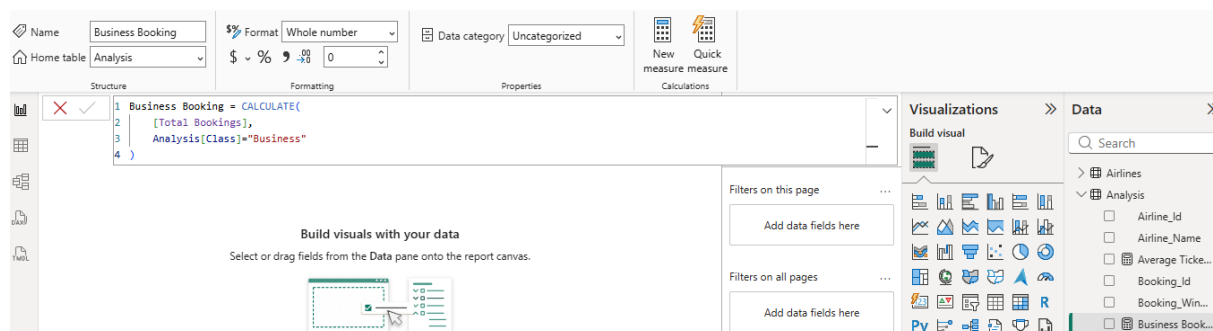
```
CALCULATE(
    [Total Bookings],
    Analysis[Class] = "Economy"
)
```



Business Bookings

Business Bookings =

```
CALCULATE(
    [Total Bookings],
    Analysis[Class] = "Business"
)
```



Calculated Fields

The Analysis table contains analytical fields such as:

- Booking Window
- Duration Category
- Price Category
- Route

These fields were created during Excel data preparation and used in Power BI for visualization and filtering.

8. Analysis and Visualizations (Power BI)

Dashboard Features

The Power BI report consists of three interactive dashboard pages designed to analyze flight booking performance, airline performance, route patterns, pricing, and customer booking behaviour.

Page 1: Flight Booking Overview

The first dashboard provides an overall summary of the flight booking dataset.

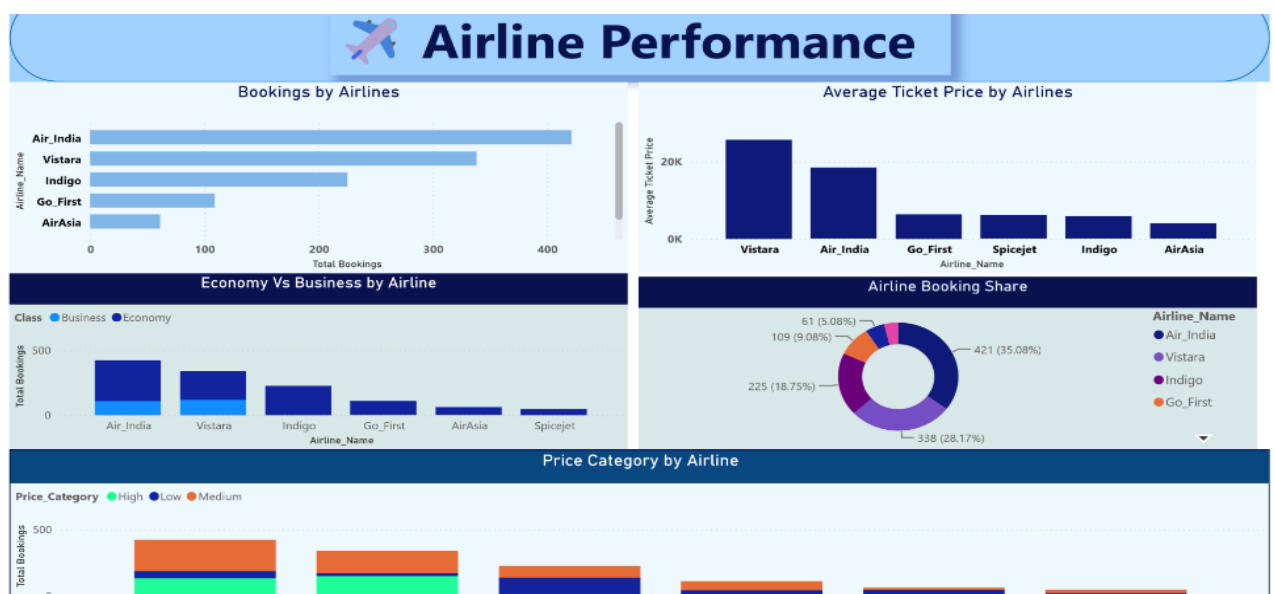
- KPI Cards: Displays Total Bookings, Average Ticket Price, Economy Bookings, and Business Bookings.
- Airline Booking Analysis: Compares booking volume across airlines.
- Class Analysis: Shows the distribution of Economy and Business bookings.
- Booking Window Analysis: Examines booking patterns based on how many days before travel the booking was made.
- Top Route Analysis: Highlights the most frequently booked routes.
- Interactive Slicers: Allows filtering by airline, class, source city, destination city, and booking window.



Page 2: Airline Performance

The second dashboard focuses on airline-level performance.

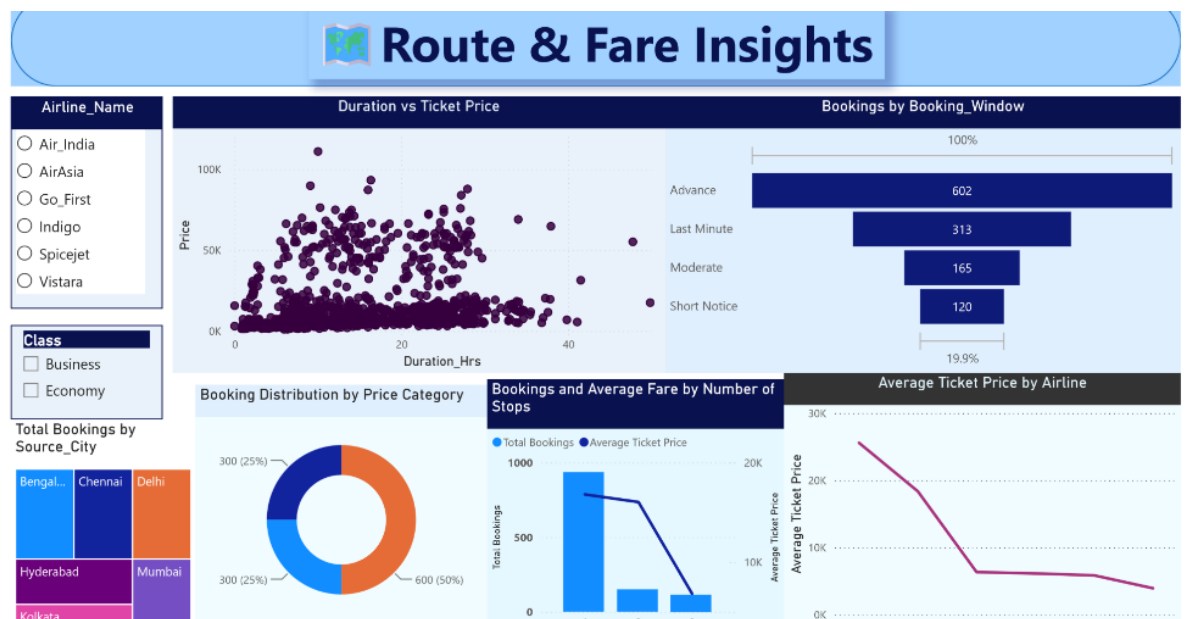
- **Airline Booking Performance:** Compares total bookings across airlines.
- **Average Ticket Price:** Compares average ticket prices between airlines.
- **Economy vs Business:** Shows class distribution across airlines.
- **Airline Booking Share:** Displays each airline's contribution to total bookings.
- **Price Category Analysis:** Compares Low, Medium, and High price bookings across airlines.
- **Interactive Filters:** Allows analysis based on airline, class, and price category.



Page 3: Route & Fare Insights

The third dashboard focuses on flight characteristics, pricing, and booking behaviour.

- **Duration vs Ticket Price:** Uses a scatter chart to examine the relationship between flight duration and ticket price.
- **Booking Window Analysis:** Uses a funnel chart to compare different booking windows.
- **Price Category Analysis:** Shows the distribution of bookings across Low, Medium, and High price categories.
- **Stops Analysis:** Compares booking volume and average ticket price based on the number of stops.
- **Duration Category Analysis:** Shows bookings across Short, Medium, and Long duration categories.
- **Airline Fare Analysis:** Compares average ticket prices across airlines.
- **Interactive Slicers:** Provides filters for airline, class, source city, and destination city.



9. Insights & Conclusions

Key Findings

- The cleaned dataset contains 1,200 valid booking records.
- Economy class represents the majority of bookings, with 977 bookings, while Business class accounts for 223 bookings.
- Air_India has the highest booking volume, with 421 bookings.
- Vistara has the second-highest booking volume, with 338 bookings.
- Indigo records 225 bookings, followed by Go_First with 109 bookings.
- AirAsia and Spicejet have comparatively lower booking volumes in the dataset.
- Vistara has the highest average ticket price, at approximately ₹25,590.52.
- Air_India has an average ticket price of approximately ₹18,373.68.
- AirAsia has the lowest average ticket price among the six airlines, at approximately ₹3,930.52.
- All 30 routes contain 40 bookings each, indicating an evenly distributed booking structure across routes.
- Ticket prices show considerable variation across airlines and travel classes.
- The analysis of booking windows helps identify whether customers tend to book closer to the travel date or further in advance.
- Flight duration and number of stops provide additional factors for understanding ticket price differences.

Descriptive Analysis

- Booking demand is concentrated among a few major airlines, particularly Air_India and Vistara.
 - Economy travel is substantially more common than Business travel in the dataset.
 - Ticket prices vary considerably between airlines.
 - Route-level booking counts are uniform, with each route containing 40 bookings, making price, duration, airline coverage, and class distribution more useful for route comparison.
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Diagnosis Analysis

- The strong concentration of bookings among certain airlines indicates differences in customer preference or airline availability within the dataset.
 - The dominance of Economy bookings suggests that the majority of customers in the dataset prefer the lower-cost travel class.
 - Large differences in average ticket prices indicate that airline and travel class are important factors affecting fare levels.
 - Since route booking volume is equal across all routes, differences in route performance should be evaluated using fare, duration, flight availability, and airline coverage rather than booking count alone.
-

Predictive Analysis

- Airlines with consistently higher booking volumes may continue to represent stronger customer demand if the observed pattern remains stable.
 - Higher-priced airlines may continue to attract customers who are willing to pay more for Business class or other service-related factors.
 - Longer-duration or multi-stop flights may show different pricing behaviour compared with shorter or direct flights.
 - Booking-window patterns can potentially help airlines understand customer planning behaviour and support better pricing strategies.
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Prescriptive Analysis

- Airlines can monitor booking-window patterns to improve pricing and seat availability strategies.
- Airlines with lower booking volumes can evaluate pricing and route-level performance to identify potential opportunities.
- Fare differences across travel classes can be used to support class-specific pricing and promotional strategies.
- Flight duration and stop patterns can be considered when evaluating ticket pricing and customer preferences.
- Since route booking volume is evenly distributed, airlines should focus on price, duration, competition, and class mix when evaluating route performance.

10. Conclusions

The Flight Booking & Travel Data Analysis project provides an overall view of airline performance, booking behaviour, route characteristics, travel classes, flight duration, stops, and ticket pricing.

The project transformed raw flight booking data into a cleaned and analysis-ready dataset using Excel and Power Query. Power BI was then used to build relationships, create DAX measures, and develop an interactive three-page dashboard.

The analysis highlights differences in airline booking volumes, Economy and Business class preferences, ticket prices, booking windows, flight duration, and stop patterns. These insights can support data-driven decisions related to airline performance, pricing, route planning, and customer travel behaviour.
